

HOMEWORK 5

Track 2

Due date: November 8, 2023.

Reading assignment: Finish reading chapter 4 of the class notes (on class web-site) and start reading chapter 5. Finish reading chapter 5 of the text book. Start reading chapters 6 & 7 of the text book. Remember to include the self-assessment grades on the first sheet of your home work (see handout for details).

1. Let X and Y be two full column-rank matrices. Show that $\text{rank}(X G Y^T) = \text{rank}(G)$. Note that X and Y might not have the same rank.
2. A complex matrix A is said to be normal if $A^H A = A A^H$. Show that a normal triangular matrix is diagonal.
3. A real matrix P is said to be an orthogonal projection if it is symmetric ($P^T = P$), and idempotent ($P^2 = P$).
 - A) What are the singular values of an orthogonal projection?
 - B) Show that there exists a full column-rank matrix U with orthonormal columns such that $P = U U^T$.
 - C) Show that $I - 2P$ is an orthogonal matrix.
4. **QR factorization with column-pivoting.** Let A be a real $m \times n$ matrix. Show that there exists an orthogonal matrix Q and a permutation matrix P such that

$$Q^T A P = \begin{pmatrix} R_{11} & R_{12} \\ 0 & 0 \end{pmatrix},$$

where R_{11} is an upper-triangular invertible matrix. From this decomposition extract an orthonormal basis for the range-space of A . How is the size of R_{11} related to the rank of A ?

5. Let A be a full column rank matrix. Show that there exists an orthogonal matrix Q and an invertible upper-triangular matrix R such that

$$A = Q \begin{pmatrix} R \\ 0 \end{pmatrix}.$$

Show how you can use the above factorization to solve the full column rank least-squares problem

$$\min_x \|A x - b\|_2.$$

6. Find the shortest distance between two infinitely straight lines in $\mathbb{R}^n$. Note, the lines do not necessarily pass through the origin. Provide the simplest expression possible. *Hint:* If $\|x\|_2 = \|y\|_2$, then $x + y$ is orthogonal to $x - y$, and the latter two lie in the span of x and y . If you use this hint, you must prove it.
7. Find the singular values of the matrix

$$C = \begin{pmatrix} X \\ I \end{pmatrix}$$

in terms of the singular values of X .

8. Find the singular values of the matrix

$$B = \begin{pmatrix} I & X \\ 0 & I \end{pmatrix}$$

in terms of the singular values of the matrix X .

9. Let σ_i denote the singular values of the matrix A . When $\sigma_k > 0$, show that

$$\min_{\text{rank}(B)=k} \|A - B\|_2 = \sigma_{k+1}.$$

Hint: Let v_i denote the right singular vectors of A . Pick a vector $x \neq 0$ in $N(B) \cap \text{Span}\{v_1, \dots, v_{k+1}\}$ and find a lower-bound on $\|(A - B)x\|_2 / \|x\|_2$. (You need to show that such an x exists!)